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ON THE GLOBAL DYNAMICS OF METAIMPACTOR LATTICES USING PHYSICS–ENHANCED MACHINE LEARNING

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Abstract. We introduce the metaimpactor, a metamaterial lattice that leverages impact dampers for enhanced energy dissipation and vibration attenuation. While discontinuous dynamics can be approximated by high-order smooth functions, they fall short in capturing the chaotic behavior and bifurcations of the metaimpactor. To address this, we develop a physics-enhanced machine learning (PEML) framework using sequential Physics-Informed Neural Networks (PINNs). Impact times—unknowns in conventional analysis—are treated as learnable variables, while the dynamics between impacts are solved via PINNs. By embedding physical constraints, the method accurately reconstructs complex, multi-impact responses. We validate the approach on a single-degree-of-freedom system, demonstrating its ability to capture nonsmooth dynamics and sequential impacts. This work lays the foundation for new classes of vibration control devices, with future extensions targeting nonlinear lattices and high-frequency, multi-impact regimes.

Key words: Nonlinear Metamaterials, Vibro–Impact, Physics-Enhanced Machine Learning

1 INTRODUCTION

The mitigation of structural vibrations under dynamic loads, such as earthquakes, remains a critical engineering challenge. Traditional passive dampers, including tuned mass dampers (TMDs) [1, 2], are constrained by narrow operational bandwidths, while active systems [3, 4] face prohibitive costs and energy demands. Recent advancements in metamaterials—artificially

structured systems with tailored wave-filtering properties—offer a promising alternative. By leveraging periodic arrangements of resonant unit cells, these systems exhibit frequency bandgaps where vibrational energy is attenuated [5, 6]. However, conventional linear metamaterials often suffer from limited bandgap widths and high cutoff frequencies, restricting their applicability to low-frequency structural vibrations.

To address these limitations, nonlinearity has been explored as part of the unit cell design, with solutions including nonlinear resonators, and impact dampers [7, 8]. Impact dampers dissipate energy through momentum exchange during impacts, enabling broadband vibration suppression [9, 10, 11]. While their effectiveness in single-degree-of-freedom systems is well-documented [12, 13], extending these benefits to metamaterial lattices poses significant modeling challenges. The nonsmooth dynamics of impacts, characterized by discontinuous velocity resets and chaotic regimes [14, 15, 16, 17], defy conventional analytical and numerical methods. Traditional solvers, such as Runge-Kutta with event detection, struggle with scalability for large lattices and often require a priori assumptions about impact timing, leading to inaccuracies in predicting bandgaps and energy dissipation.

Modeling the complex, nonsmooth behavior of impact-based systems requires methods that can capture both continuous dynamics and discontinuous events in a unified framework. Physics-Informed Neural Networks (PINNs) offer a promising direction in this regard. By embedding the governing equations and boundary or initial conditions directly into the loss function of a neural network, PINNs enable the solution of forward and inverse problems [18, 19, 20]. This approach has proven effective across diverse applications, including fluid mechanics [21], material modeling [22], contact mechanics [23], nonsmooth dynamical systems [24], and structural dynamics [25]. The use of PINNs for systems with repeated impacts and discontinuous velocity profiles remains a challenging and relatively unexplored area. However, decomposing the learning paradigm into simpler domains has been shown to benefit PINNs; such as with parallel [26] or sequential/recurrent [27] PINNS which reduce the scope of individual training domains.

This work introduces a sequential Physics-Informed Neural Network (PINN) framework for modeling the nonsmooth dynamics of vibro-impact systems with internal impact dampers. Impact times are treated as learnable variables, enabling the network to identify discontinuities directly from the governing physics. Each PINN segment models the response between successive impacts and enforces the dynamics and impact conditions during training. Upon impact, velocities are updated using momentum and restitution laws, and the resulting state initializes the next segment. Chaining these segments yields a complete multi-impact response without requiring explicit event detection. While the framework is designed for nonlinear lattice systems, this study focuses on a single-degree-of-freedom (SDOF) configuration to demonstrate its core mechanisms. This simplified case enables transparent validation and demonstrates the framework's ability to capture nonsmooth dynamics and sequentially identify multiple impacts based on embedded physical constraints. The proposed framework will be further developed for application to full-scale metamaterial lattices, facilitating dispersion analysis and performance validation under high-frequency, multi-impact conditions.

2 PROBLEM DESCRIPTION AND GOVERNING EQUATIONS

2.1 Nonlinear Meta-Impact Lattice

The system under consideration is a nonlinear metamaterial lattice composed of a consecutive arrangement of unit cells, each incorporating an impact damper. The primary objective of this configuration is to reduce the vibrational response of a protected structure through energy dissipation induced by internal impacts. Each unit cell consists of a rigid container of mass m, connected to neighboring cells via springs with stiffness k and damping coefficient c, and housing a free internal mass μ that can translate laterally. The internal mass is allowed to move within a bounded region defined by a clearance D on either side and interacts with the container walls through impact events.

The unit cells are arranged in a chain and elastically connected to each other. The leftmost cell is fixed, while the rightmost cell is attached to a primary oscillator with mass M, stiffness K, and damping C. An external force F is applied to the primary mass. This configuration is illustrated in Figure 1.

Dynamic nonlinearity arises when the relative displacement between the outer mass and the inner mass in a unit cell reaches the clearance limit. At such instances, a discontinuous change in velocity occurs due to impact, introducing complex nonsmooth dynamics that are challenging to model using classical continuous-time approaches.

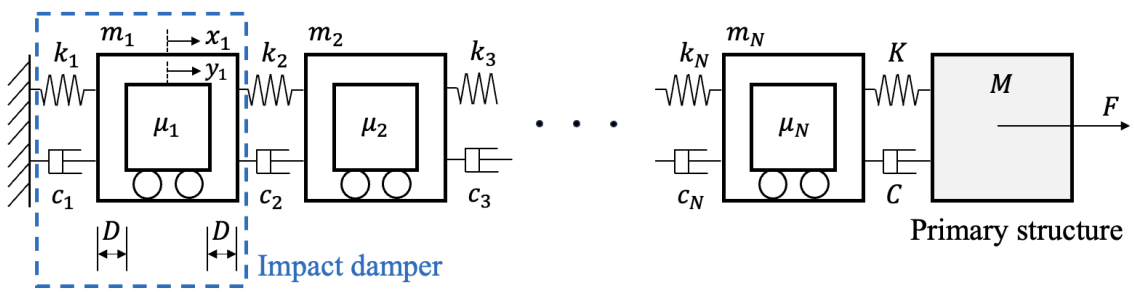


Figure 1: Nonlinear lattice of impact dampers coupled to a primary structure.

2.2 Governing Equations

The global equation of motion for the nonlinear metamaterial lattice coupled to a primary structure can be expressed as:

$$\mathbf{M}\ddot{\mathbf{X}}(t) + \mathbf{C}\dot{\mathbf{X}}(t) + \mathbf{K}\mathbf{X}(t) = \mathbf{F}(t), \tag{1}$$

where $\mathbf{X}(t) \in \mathbb{R}^{N+1}$ is the vector of nodal displacements of the N unit cells and the main structure:

$$\mathbf{X}(t) = \begin{bmatrix} x_1(t) & x_2(t) & \cdots & x_N(t) & X(t) \end{bmatrix}^\top. \tag{2}$$

Here, $x_i(t)$ denotes the displacement of the i-th container mass in the lattice, and $X(t)$ is the displacement of the protected primary mass. The external excitation $\mathbf{F}(t)$ is nonzero only in the last component:

$$\mathbf{F}(t) = \begin{bmatrix} 0 & \cdots & 0 & F(t) \end{bmatrix}^\top. \tag{3}$$

The mass matrix $\mathbf{M}$ is diagonal and expressed as:

$$\mathbf{M} = \text{diag}(m_1, m_2, \ldots, m_N, M). \tag{4}$$

The stiffness matrix $\mathbf{K}$ is tridiagonal and expressed as:

$$\mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 & 0 & \cdots & 0 & 0 \\ -k_2 & k_2 + k_3 & -k_3 & \cdots & 0 & 0 \\ 0 & -k_3 & k_3 + k_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & k_N + K & -K \\ 0 & 0 & 0 & \cdots & -K & K \end{bmatrix}. \tag{5}$$

The damping matrix $\mathbf{C}$ is tridiagonal and expressed as:

$$\mathbf{C} = \begin{bmatrix} c_1 + c_2 & -c_2 & 0 & \cdots & 0 & 0 \\ -c_2 & c_2 + c_3 & -c_3 & \cdots & 0 & 0 \\ 0 & -c_3 & c_3 + c_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & c_N + C & -C \\ 0 & 0 & 0 & \cdots & -C & C \end{bmatrix}. \tag{6}$$

Each internal mass $y_i(t)$ evolves as:

$$\mu_i \ddot{y}_i(t) = 0, \quad \text{subject to} \quad |x_i(t) - y_i(t)| < D. \tag{7}$$

This formulation governs the structure between impact events. Discontinuities due to impact are handled by conservation laws, detailed in the subsequent sections.

2.3 Impact Conditions

The impact condition is enforced when the displacement gap reaches the clearance:

$$|x_i(t) - y_i(t)| = D, \tag{8}$$

triggering a discontinuous velocity change governed by momentum and restitution laws.

To characterize the discontinuous jump in velocity at impact, we introduce the coefficient of restitution $r \in [0, 1]$, which defines the ratio of relative velocities after and before impact [12]:

$$r = -\frac{\dot{y}_i^+ - \dot{x}_i^+}{\dot{y}_i^- - \dot{x}_i^-}, \tag{9}$$

where $\dot{x}_i^-$ and $\dot{y}_i^-$ are the pre-impact velocities, and $\dot{x}_i^+$, $\dot{y}_i^+$ are the post-impact velocities.

Momentum conservation during the (assumed instantaneous) impact yields:

$$m_i\dot{x}_i^+ + \mu_i\dot{y}_i^+ = m_i\dot{x}_i^- + \mu_i\dot{y}_i^-. \tag{10}$$

Combining Eqs. (9) and (10), the post-impact velocities can be computed using:

$$\begin{bmatrix}\dot{x}_i^+\\ \dot{y}_i^+\end{bmatrix} = \frac{1}{m_i+\mu_i}\begin{bmatrix} m_i - r\mu_i & (1+r)\mu_i \\ (1+r)m_i & \mu_i - rm_i\end{bmatrix}\begin{bmatrix}\dot{x}_i^-\\ \dot{y}_i^-\end{bmatrix}. \tag{11}$$

2.4 Modeling Challenges and Motivations

Modeling impact-based systems poses significant challenges due to the presence of nonsmooth dynamics and discontinuous velocity changes. Traditional event-based solvers require precise impact detection and manual velocity resets, which can become cumbersome and numerically sensitive, especially in systems with multiple impacts or chaotic behavior.

To overcome these limitations, the problem is reformulated using a sequential Physics-Informed Neural Network (PINN) framework that incorporates both the governing equations and impact conditions directly into the training process. The architecture of the proposed sequential PINN framework is detailed in the following section.

3 SEQUENTIAL PHYSICS-INFORMED NEURAL NETWORK FRAMEWORK

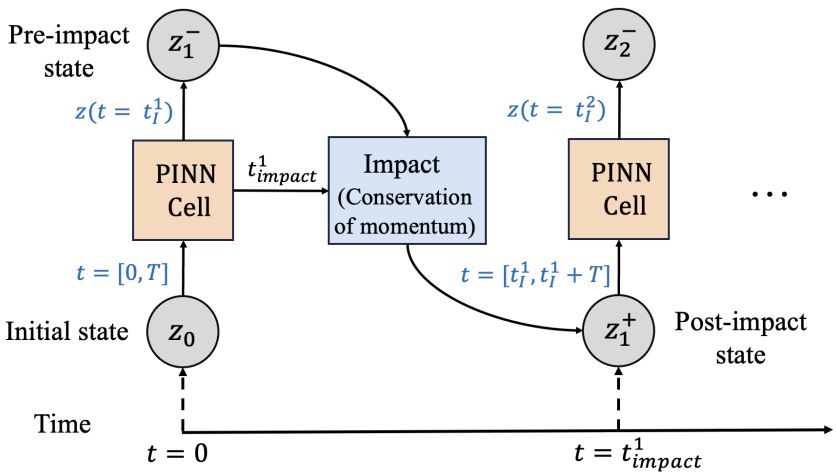
3.1 Overview of the Framework

While the proposed sequential PINN framework is formulated with the general goal of modeling nonlinear lattice systems with internal impact dampers, its structure is equally well-suited to single-unit configurations. In this work, a single-degree-of-freedom system is used to demonstrate the key features of the framework—namely, its ability to identify impact times, model nonsmooth dynamics, and reconstruct the system response through modular PINN segments. An overview of the sequential framework is shown in Figure 2.

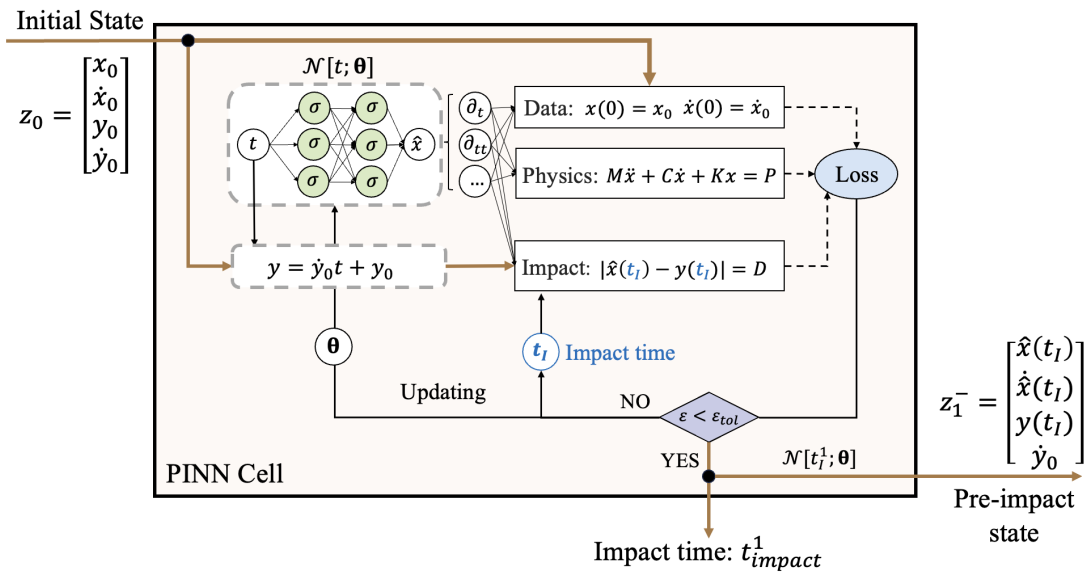
As shown in Figure 2(a), each PINN cell models the system response between two successive impacts. Figure 2(b) illustrates the internal structure of a PINN cell, at the beginning of each segment, the system state $\mathbf{z}_0 = [x_0, \dot{x}_0, y_0, \dot{y}_0]^\top$ is used as the initial condition. The network takes time $t \in [0, T]$ as input and predicts the displacement of the primary mass $x(t)$. The displacement of the internal mass is defined analytically, based on its initial displacement and velocity, as

$$y(t) = y_0 + \dot{y}_0 t. \tag{12}$$

The neural network $x(t;\theta)$ consists of fully connected layers with `tanh` activation functions, chosen to model smooth dynamics between impacts. The first and second order time derivatives of $x(t)$ are computed using automatic differentiation. The impact time t_{impact} is treated as a learnable scalar variable, optimized during training to satisfy the impact constraint. When an impact occurs, post-impact velocities are computed based on momentum conservation and restitution laws, and the resulting state is passed forward to initialize the next PINN cell.



(a) Proposed Sequential PINN framework for modeling impact dampers.



(b) PINN Cell.

Figure 2: (a) Proposed Sequential PINN framework for modeling impact dampers; and (b) PINN Cell.

3.2 Physics-Informed Loss Function

Each PINN cell is trained by minimizing a composite loss function that enforces physical consistency of the solution within its time interval. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \beta_{\text{IC}}\mathcal{L}_{\text{IC}} + \beta_{\text{ODE}}\mathcal{L}_{\text{ODE}} + \beta_{\text{impact}}\mathcal{L}_{\text{impact}}, \tag{13}$$

where β_{IC}, β_{ODE}, and β_{impact} are weights balancing the convergence rate of each loss term.

The first term, the initial condition loss $\mathcal{L}_{\text{IC}}$, ensures agreement with the known initial displacement and velocity of the primary mass at the start of the interval:

$$\mathcal{L}_{\text{IC}} = (x(0) - x_0)^2 + (\dot{x}(0) - \dot{x}_0)^2 . \tag{14}$$

The second term, the ODE residual loss $\mathcal{L}_{\text{ODE}}$, penalizes violations of the governing equation over a set of collocation points $\{t_i\}_{i=1}^{N_r}$:

$$\mathcal{L}_{\text{ODE}} = \frac{1}{N_r} \sum_{i=1}^{N_r} (m\ddot{x}(t_i) + c\dot{x}(t_i) + kx(t_i))^2 . \tag{15}$$

The final term, the impact condition loss $\mathcal{L}_{\text{impact}}$, enforces the occurrence of an impact at an unknown time $t = t_{\text{impact}}$. This is achieved by penalizing deviations between the predicted relative displacement and the known clearance D at that instant:

$$\mathcal{L}_{\text{impact}} = (|x(t_{\text{impact}}) - y(t_{\text{impact}})| - D)^2 . \tag{16}$$

3.3 Determination of the Impact Time and Estimation of the Structural Response

The training procedure minimizes the total loss by jointly optimizing the neural network parameters θ and the impact time variable t_{impact} within each PINN cell. At each training iteration, the network is evaluated at both the collocation points and the current estimate of t_{impact}. The total loss is then computed and backpropagated to update both θ and t_{impact}.

Once trained, the network identifies the corresponding impact time and predicts the displacement $x(t)$, along with its first and second derivatives $\dot{x}(t)$ and $\ddot{x}(t)$. At the estimated impact time, the velocities of the primary and internal masses are updated using conservation of momentum and restitution laws. The resulting post-impact state serves as the initial condition for the subsequent PINN cell.

This procedure is repeated for each impact segment using a sequence of independently trained PINN cells. The full multi-impact response of the system is reconstructed by concatenating the outputs of all segments. To improve convergence and stability, a two-stage training strategy is employed: the loss function is initially minimized using the Adam optimizer with a predefined stopping condition, followed by refinement using the L-BFGS-B algorithm. Pseudocode outlining the proposed sequential PINN framework is provided in Algorithm 1.

4 NUMERICAL RESULTS

4.1 Simulation Setup

To evaluate the performance of the proposed framework, a single impact damper unit cell is considered as a representative test case. The system parameters are defined as follows: primary mass $m = 1\,\text{kg}$, internal mass $\mu = 0.3\,\text{kg}$, spring stiffness $k = 1\,\text{N/m}$, damping coefficient $c = 0\,\text{Ns/m}$, and clearance distance $D = 1\,\text{m}$. The total simulation time is set to $T = 20\,\text{s}$.

The initial conditions are given by $x(0) = 0\,\text{m}$, $\dot{x}(0) = 0\,\text{m/s}$, $y(0) = 0\,\text{m}$, and $\dot{y}(0) = -1\,\text{m/s}$. Both the sequential PINN framework and the event-based reference simulation are initialized with these same conditions to ensure consistency and fairness in the comparison.

Algorithm 1 Sequential PINN Framework for Impact Time and Response Estimation

Require: Initial state $\mathbf{z}_0 = [x_0, \dot{x}_0, y_0, \dot{y}_0]$, parameters m, c, k, μ, D, either number of impacts N_{imp} or final time T

Set $j = 1$, $t_0 = 0$
while (not reached T) **or** (not reached N_{imp}) **do**
  Initialize PINN cell with parameters $\theta^{(j)}$ and learnable $t_{\text{impact}}^{(j)}$
  Sample collocation points in $[t_{j-1}, T]$
  Define loss with initial condition, governing equation, and impact constraints
  Minimize loss using Adam followed by L-BFGS-B
  Estimate $t_{\text{impact}}^{(j)}$ and predict $x^{(j)}(t)$, $\dot{x}^{(j)}(t)$, and $\ddot{x}^{(j)}(t)$ in $[t_{j-1}, t_{\text{impact}}^{(j)}]$
  Apply impact law; update state $\mathbf{z}_j$, set $t_j = t_{\text{impact}}^{(j)}$
  Increment $j \leftarrow j + 1$
end while
return Estimated impact times and full response $x(t)$, $\dot{x}(t)$, $\ddot{x}(t)$, $y(t)$, $\dot{y}(t)$

4.2 Determination of the Impact Times

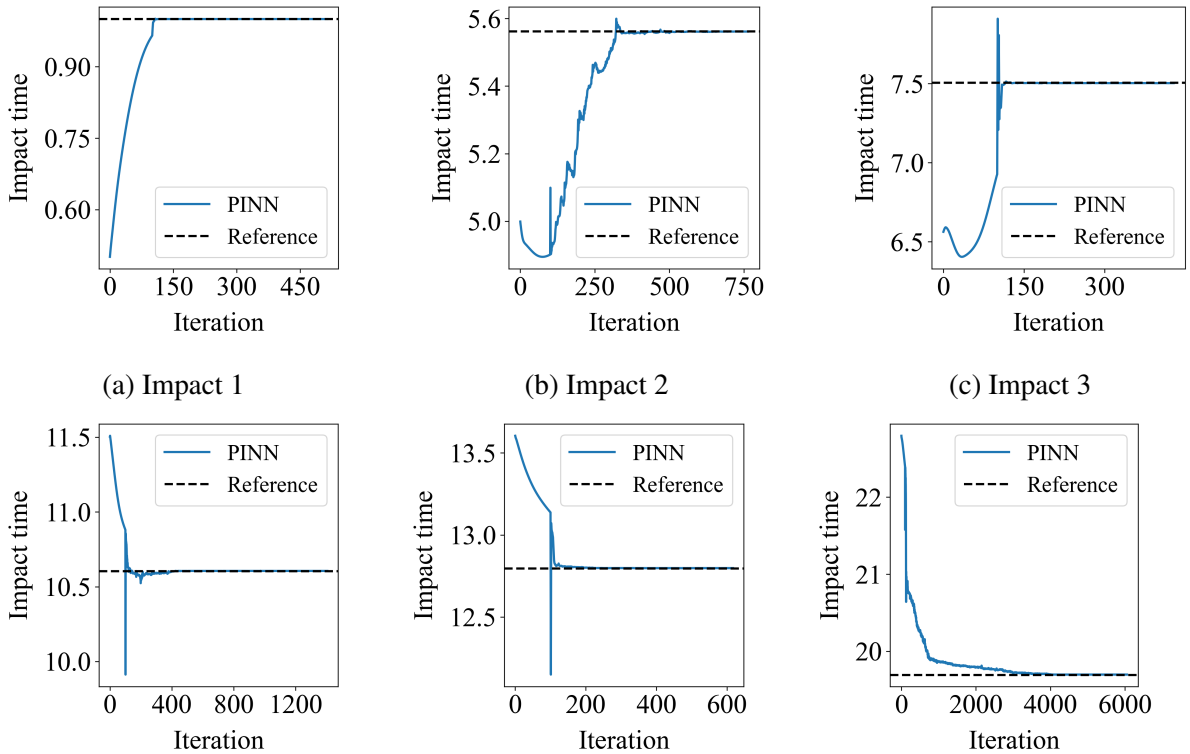
A key feature of the proposed sequential PINN framework is its ability to discover the impact times through physics-constrained optimization. During training, each impact time $t_{\text{impact}}^{(i)}$ is treated as a learnable scalar variable. The corresponding loss term $\mathcal{L}_{\text{impact}}^{(i)}$ is minimized jointly with the residuals of the governing equations and initial conditions.

Figure 3 illustrates the training convergence of impact time estimates across six successive impacts. Each subplot shows the evolution of $t_{\text{impact}}^{(i)}$ over training iterations. Dashed horizontal lines indicate the reference values computed by the event-based solver. The results demonstrate that the PINN framework reliably converges to accurate impact times that closely match those obtained from the event-based solver.

4.3 Structural Response Estimation

Figures 4(a)–(d) present the structural response predicted by the sequential PINN framework. In all panels, vertical dashed lines indicate the estimated impact times. Figure 4(a) shows the displacement of the primary and internal masses, while Figure 4(b) presents their corresponding velocities. Figure 4(c) plots the relative displacement $x(t) - y(t)$, which captures the onset of impacts as it approaches the clearance threshold. Figure 4(d) illustrates the impulse history, reflecting the momentum exchange resulting from the impacts.

For comparison, impact times are first determined using an event-based method, and the structural response between impacts is then computed analytically from the closed-form solution of the undamped system. As shown in Figure 4(c), the results demonstrate that the sequential PINN framework can accurately reconstruct both the smooth dynamics and nonsmooth impact events of the system.



(a) Impact 1 (b) Impact 2 (c) Impact 3

(d) Impact 4 (e) Impact 5 (f) Impact 6

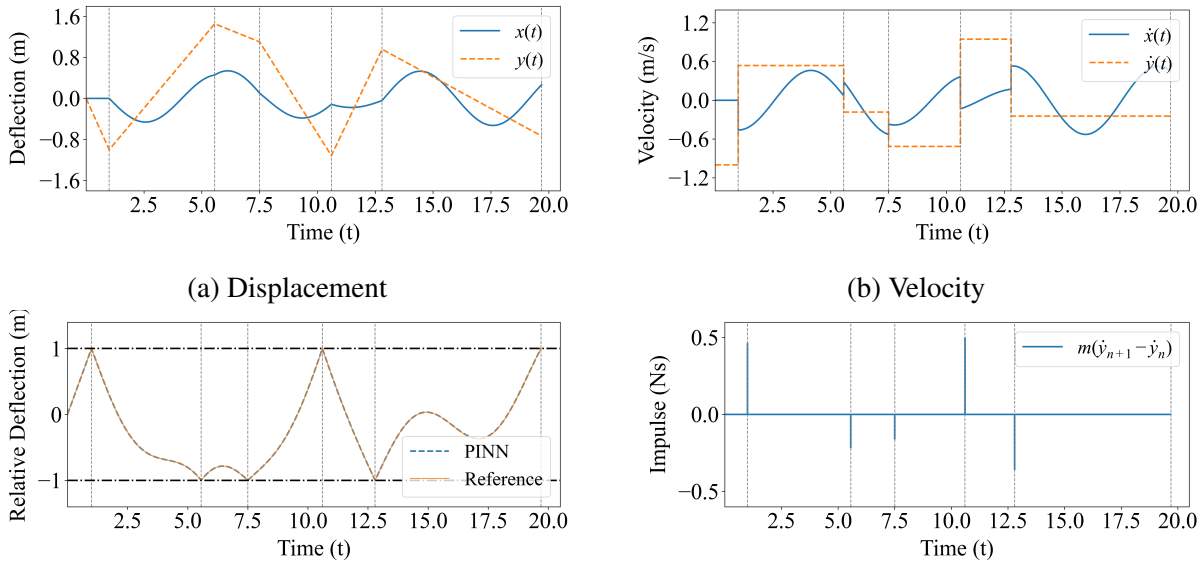
Figure 3: Training convergence of impact time estimation using PINN.

5 CONCLUSION

This work presents a sequential Physics-Informed Neural Network (PINN) framework for modeling the dynamics of vibro-impact systems equipped with internal impact dampers. By embedding both the governing equations and impact constraints into the loss function, the method enables direct learning of nonsmooth dynamics from the underlying physics, without relying on labeled data or explicit event detection. A key feature of the framework is the treatment of impact times as learnable variables, allowing the network to identify discontinuities through physics-constrained optimization.

The structural response is captured by linking multiple PINN segments, each responsible for modeling the system behavior between two consecutive impacts. At each predicted impact, momentum-based velocity updates are applied, and the resulting post-impact state is used to initialize the next segment. This recursive structure provides a modular and physically consistent approach to modeling repeated impact events.

The framework has been validated on a single degree-of-freedom impact damper system. The results demonstrate that the PINN accurately identifies both the impact times and the dynamic response of the system. Future developments will target the extension of this framework



(a) Displacement

(b) Velocity

(c) Relative displacement

(d) Impulse time history

Figure 4: Structural response of the impact damper system predicted by the sequential PINN framework. (a) displacement and (b) velocity of primary and internal masses, (c) relative displacement compared to the reference impact threshold, and (d) impulse time history. Vertical dashed lines indicate impact events.

to lattice systems with many unit cells and more frequent impact events.

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